



CV2090
Probability & Statistics for
Civil Engineers
(Jul – Nov 2026)

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L13
Aug 20, 2026

Announcement

- 24th Aug:
 - Lecture hour 3 – 4 pm
 - Tutorial 4 – 5 pm
- 25th Aug: No class on Tuesday
- 31st Aug: Quiz-1 (8 AM)
- Check course page on GitHub regularly

<https://github.com/prakashvs613/2026-CV2090-IITM/>

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Messages from the Last Class

- **Message 1:** Higher moments

Expectation of $g(x)$, $E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x) dx$ LOTUS

m^{th} moment, $g(x) \equiv x^m$
 $E[x^m] = \int_{-\infty}^{\infty} x^m f_x(x) dx$

m^{th} central moment, $g(x) = (x - \mu)^m$
 $\rightarrow 1^{\text{st}}$ moment, $E[x] = \mu$
 $\rightarrow 2^{\text{nd}}$ moment, $E[x^2]$

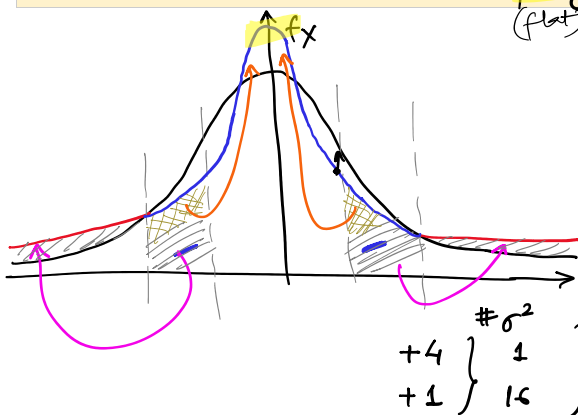
1^{st} central moment, $\rightarrow 0 = E[(x - \mu)]$

2^{nd} central moment, $\rightarrow \sigma^2 = E[(x - \mu)^2] = E[x^2] - \mu^2$

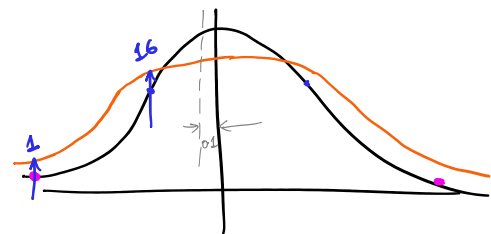
3^{rd} central moment, $\rightarrow$ scaled skewness $\rightarrow$ skewness $= \frac{E[(x - \mu)^3]}{\sigma^3}$

4^{th} central moment $\rightarrow k = \frac{E[(x - \mu)^4]}{\sigma^4}$
 $\rightarrow \theta > 0$: Right/+ve skewness
 $\rightarrow \theta < 0$: Left/-ve skewness
 (sharp) $k > 3$ leptokurtic
 (medium) $k = 3$ Mesokurtic
 (flat) $k < 3$ platykurtic

excess kurtosis = $k - 3$,



- $\rightarrow$ Centrality μ
- $\rightarrow$ Dispersion σ^2
- $\rightarrow$ Skewness θ
- $\rightarrow$ Tailedness/ k
- ~~peakedness~~
NOT About peaks



Messages from the Last Class

- **Message 2:** Summary Statistics $\neq$ Distribution

Data sources Dozen

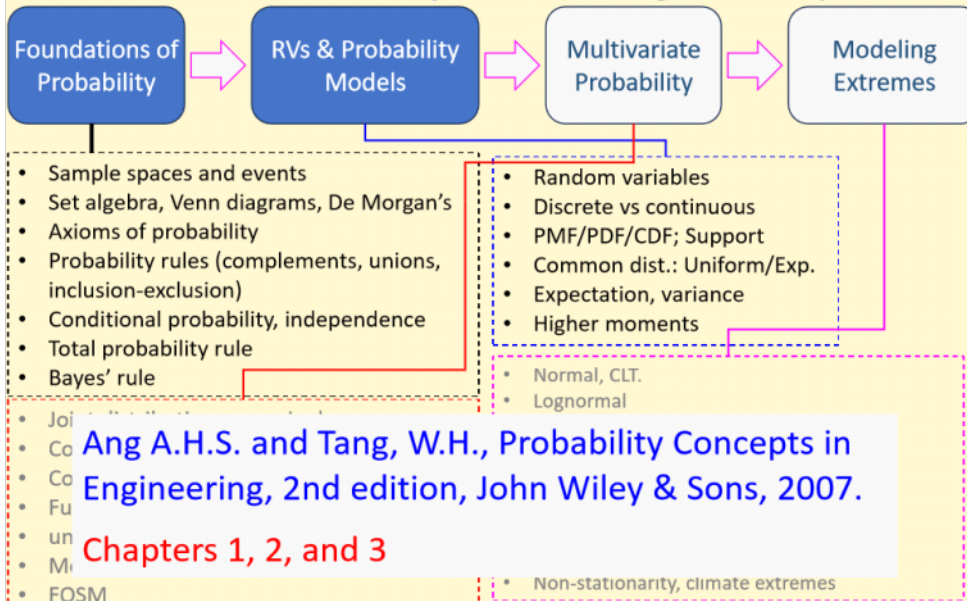
- 12 datasets with exactly equal μ, σ^2
- "Dino" data has same μ, σ^2

eyeball the data.

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Module I: Chapter 2 (M1C2)

Module 1: **Probability** – Understanding Uncertainty



Functions of Single rv

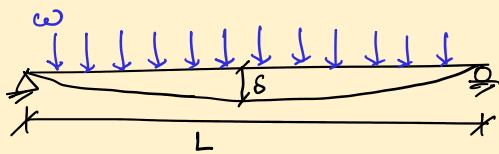
Ang A.H.S. and Tang (Section 4.2.1)

Why do we need functions of RV?

* wind speed $P_d = \frac{1}{2} \rho V^2$

$$V \longrightarrow P_d$$

* Beam



$$\delta = \frac{w L^4}{384 E I}$$

* Energy released in an Earthquake of magnitude M ,
 $E = 10^{1.5M}$

$$E_7/E_6 \approx 32$$

Why do we need functions of RV?

→ functions of 'observed RV' have physical significance.

$$g(x) = 2x + 3$$

$$E[g(x)] = 2 \times E[x] + 3$$

$$\sigma^2[g(x)] = 2^2 \times \sigma_x^2$$

$$\Rightarrow \sigma[g(x)] = 2 \times \sigma_x$$

$$g(x) = x^2$$

$$E[g(x)] \stackrel{?}{=} g[E(x)]$$

$$g(x) = x^3 + 5 \sin x + e^x$$

$$E[g(x)] = E[x^3] + 5E[\sin x] + E[e^x]$$

Expectation of Functions

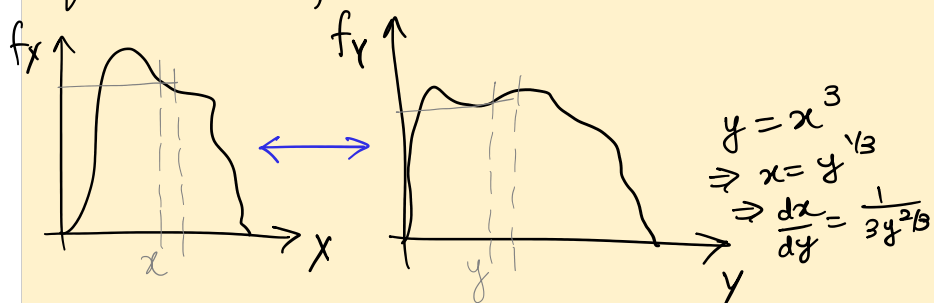
Expectation is a linear operator

E & $g(x)$ are commutative, if $g(x)$ is
an affine transformation
 $ax + b$

$$E[g(x)] = g[E(x)]$$

Function of Single RV

Let's say, X has a pdf $f_X(x)$
find PDF for $Y = X^3$.



$$f_X dx = f_Y dy$$

$$\Rightarrow f_Y = f_X \left| \frac{dx}{dy} \right| = f_X(y^{1/3}) \cdot \left| \frac{1}{3y^{2/3}} \right|$$

Jacobian

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Function of Single RV

$$Y = X^3$$

Given PDF of X , determine $f_Y(y)$.

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Mean and Variance propagation laws

- A random variable, $X \sim \mathcal{N}(40,5)$

If we define, another random variable, $Y = 3X + 10$, what are its mean and standard deviation (stdev) and type.

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Nonlinear Functions of Single rv